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LINEAR VIBRATION MOTOR AND ELECTRONIC DEVICE

Abstract

Provided are a linear vibration motor and an electronic device. In a receiving space enclosed by a case, a follower is at a support frame of a vibrator, and a driver is opposite to the follower for driving the follower to drive the support frame to move along a straight line. The support frame is provided with receiving grooves. Each receiving groove is provided with at least one support member, part of which is located outside the receiving groove. The support member is pressed against the case by the support frame and configured to support the support frame. When the driver drives the support frame to move relative to the case, the support member rolls relative to the case and the support frame, the support frame stably moves, and the receiving groove is located in the receiving space and does not deform due to an influence of any impact.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to the technical field of vibration motors, in particular to a linear vibration motor and an electronic device.

BACKGROUND

[0002] With the development of electronic technologies, portable electronic devices such as smart phones and handheld game consoles are becoming more and more popular. Most electronic devices use linear vibration motors as feedback devices of the system, such as incoming call prompts, information prompts, navigation prompts, and vibration feedback of game consoles.

[0003] An existing linear vibration motor usually uses a leaf spring as a support member of a vibrator, and an elastic leaf spring is an excellent device for fixing and supporting the vibrator. However, the impact resistance and vibration resistance of the leaf spring are poor, and when the leaf spring is subjected to falling impact or reciprocating motion/vibration caused by repeated vibration of the motor, the leaf spring is prone to deformation and damage, so that the linear vibration motor using the leaf spring as a support member has poor performance in falling tests and life tests, and is also prone to problems in practical applications. In view of the problems existing in the leaf spring, the leaf spring with good vibration resistance and impact resistance is usually manufactured by changing the thickness of the leaf spring or changing the material of the leaf spring, however, it is very difficult to determine the suitable thickness and manufacturing material of the leaf spring, and when the leaf spring has enough elasticity to resist impact and vibration, the strength of the leaf spring is reduced, resulting in a reduction of the supporting effect of the leaf spring, and the controllability of the leaf spring is also reduced. In addition, when the linear vibration motor is assembled, the leaf spring is prone to being bent, damaged or even broken.

[0004] Therefore, it needs to provide a linear vibration motor capable of solving the above technical problems.

SUMMARY

[0005] The present disclosure provides a linear vibration motor which has high reliability and excellent impact resistance and vibration resistance.

[0006] In an aspect, an embodiment of the present disclosure provides a linear vibration motor, including: a case having a receiving space; a vibrator including a support frame and a follower fixed to the support frame, the support frame being arranged in the receiving space, and the support frame being provided with a plurality of receiving grooves; a driver provided opposite to the follower, the driver being arranged in the receiving space, and the driver being configured to drive the follower to drive the support frame to move along a straight line; and a plurality of support members. Each of the plurality of receiving grooves is provided with at least one of the plurality of support members, a part of one of the plurality of support members is located outside the receiving groove, the plurality of support frames press the plurality of support members against the case; and when the driver drives the support frame to move relative to the case, the support member rolls relative to the case and the support frame.

[0007] As an improvement, the linear vibration motor further includes a second magnetic member arranged at the case, the driver is a coil, and the coil is arranged at the case; the follower is a first magnetic member, the second magnetic member is disposed opposite to the first magnetic member, and the second magnetic member is configured to cooperate with the first magnetic member to maintain the support frame at an initial position when the coil is powered off, and drive the first magnetic member to drive the support frame to move along the straight line when the coil is powered on; or the linear vibration motor further includes a second magnetic member arranged at the support frame, the driver is a first magnetic member, and the first magnetic member is arranged

at the case; the follower is a coil, the second magnetic member is disposed opposite to the first magnetic member, and the second magnetic member is configured to cooperate with the first magnetic member to maintain the support frame at an initial position when the coil is powered off, and the second magnetic member is further configured to be driven to drive the support frame to move along the straight line when the coil is powered on.

[0008] As an improvement, the case includes a front shell and a base, the front shell is of a groove structure, and the base covers the groove structure to enclose the receiving space with the groove structure; and the receiving groove is arranged at a side of the support frame facing the front shell and extends along the straight line, an inner wall of the receiving groove and an inner wall of the front shell clamp and fix the support member, and the support member is a ball.

[0009] As an improvement, the inner wall of the front shell includes a front wall opposite to the support frame, and a side wall surrounding the front wall and connected to the front wall; the inner wall of the receiving groove includes an inclined wall which forms an included angle with the front wall; and the ball is located between the front wall, the side wall and the inclined wall.

[0010] As an improvement, the included angle between the inclined wall and the front wall is within a range from 30° to 60°.

[0011] As an improvement, an anti-collision portion is provided at a side of the support frame away from the receiving groove, and in a direction perpendicular to the base, a width of a gap between the anti-collision portion and the base is less than a height of the support member; and the anti-collision portion is configured to prevent the support frame from colliding with the base.

[0012] As an improvement, the case includes a front shell and a base, the front shell is a groove structure, the base covers the groove structure to enclose the receiving space with the groove structure; and the receiving groove is arranged at a side of the support frame facing the base and extends along the straight line, an inner wall of the receiving groove and the base clamp and fix the support member, and the support member is a ball.

[0013] As an improvement, the front shell includes a side wall surrounding the base and connected to the base, an inner wall of the receiving groove includes an inclined wall which forms an included angle with the base, and the ball is located between the base, the side wall and the inclined wall.

[0014] As an improvement, the included angle between the inclined wall and the base is within a range from 30° to 60°.

[0015] As an improvement, an anti-collision portion is provided at a side of the support frame away from the receiving groove, and in a direction perpendicular to the base, a width of a gap between the anti-collision portion and the base is less than a height of the support member; and the anti-collision portion is configured to prevent the support frame from colliding with the front shell.

[0016] As an improvement, the second magnetic member is a magnetic yoke.

[0017] As an improvement, the second magnetic member is magnetic steel, and a magnetic pole of the second magnetic member is arranged opposite to a magnetic pole of the first magnetic member.

[0018] In another aspect, an embodiment of the present disclosure provides an electronic device, including a device body and a linear vibration motor arranged at the device body. The linear vibration motor includes: a case having a receiving space; a vibrator including a support frame and a follower fixed to the support frame, the support frame being arranged in the receiving space, and the support frame being provided with a plurality of receiving grooves; a driver provided opposite to the follower, the driver being arranged in the receiving space, and the driver being configured to drive the follower to drive the support frame to move along a straight line; and a plurality of support members. Each of the plurality of receiving grooves is provided with at least one of the plurality of support members, a part of one of the plurality of support members is located outside the receiving groove, the plurality of support frames press the plurality of support members against the case; and when the driver drives the support frame to move relative to the case, the support member rolls relative to the case and the support frame.

[0019] Compared with the solution in the related art, in the embodiments of the present disclosure,

in a receiving space enclosed by the case, a follower is arranged at a support frame of the vibrator, and the driver is disposed opposite to the follower for driving the follower to drive the support frame to move along a straight line. The support frame is provided with a plurality of receiving grooves, each receiving groove is provided with at least one support member, and a part of the support member is located outside the receiving groove. The support member is pressed against the case by the support frame, the support member is configured to support the support frame, and when the driver drives the support frame to move relative to the case, the support member rolls relative to the case and the support frame, so that the support frame can stably move, and the receiving groove is located in the receiving space and does not deform due to the influence of an impact.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0020] One or more embodiments are exemplarily described in the corresponding figures, and these exemplified descriptions do not constitute limitations on the embodiments. Elements with a same reference numeral in the figures represent similar elements, and unless otherwise stated, the figures do not represent proportional limitations.

[0021] FIG. 1 is a three-dimensional structural schematic diagram of a linear vibration motor according to Embodiment I of the present disclosure;

[0022] FIG. 2 is a schematic diagram of a front view of a linear vibration motor according to Embodiment I of the present disclosure;

[0023] FIG. 3 is a schematic diagram of a side view of a linear vibration motor according to Embodiment I of the present disclosure;

[0024] FIG. 4 is a schematic diagram of assembly of internal components of a linear vibration motor according to Embodiment I of the present disclosure;

[0025] FIG. 5 is a schematic diagram of an exploded view of a linear vibration motor according to Embodiment I of the present disclosure;

[0026] FIG. 6 is an enlarged view of a region S1 shown in FIG. 5;

[0027] FIG. 7 is a cross section along AA' shown in FIG. 2;

[0028] FIG. 8 is an enlarged view of a portion shown in FIG. 7;

[0029] FIG. 9 is a cross section along BB' shown in FIG. 2;

[0030] FIG. 10 is a cross sectional of a linear vibration motor according to Embodiment I of the present disclosure, where a second magnetic member of the linear vibration motor is a magnetic steel;

[0031] FIG. 11 is a schematic diagram of another configuration of a linear vibration motor according to Embodiment I of the present disclosure;

[0032] FIG. 12 is a schematic diagram of another configuration of a linear vibration motor according to Embodiment I of the present disclosure;

[0033] FIG. 13 is a schematic diagram of a configuration of a linear vibration motor according to Embodiment II of the present disclosure;

[0034] FIG. 14 is a schematic diagram of another configuration of a linear vibration motor according to Embodiment II of the present disclosure;

[0035] FIG. 15 is a three-dimensional structural schematic diagram of an electronic device according to Embodiment III of the present disclosure;

[0036] FIG. 16 is a three-dimensional structural schematic diagram of another electronic device according to Embodiment III of the present disclosure; and

[0037] FIG. 17 is a three-dimensional structural schematic diagram of another electronic device according to Embodiment III of the present disclosure.

REFERENCE SIGNS

[0038] **100** linear vibration motor [0039] **10** case [0040] **11** receiving space [0041] **12** front shell [0042] **13** base [0043] **20** vibrator [0044] **21** support frame [0045] **211** receiving groove [0046] **2111** inclined wall [0047] **2112** bottom wall [0048] **2113** upper wall [0049] **212** anti-collision portion [0050] **213** avoidance groove [0051] **22** follower [0052] **30** driver [0053] **40** support member [0054] **50** second magnetic member [0055] **60** circuit board [0056] **200** electronic device [0057] **210** device body

DESCRIPTION OF EMBODIMENTS

[0058] In order to better illustrate objectives, technical solutions, and advantages of embodiments of the present disclosure, the technical solutions in embodiments of the present disclosure are described in details with reference to the accompanying drawings in the following description. Those of ordinary skill in the art should understand that in the embodiments of the present disclosure, various technical details are set forth for the reader to better understand the present disclosure, but without these technical details and various variations and modifications based on the following embodiments, the technical solutions claimed in the present disclosure can still be implemented.

[0059] In the embodiments of the present disclosure, terms such as “upper”, “lower”, “left”, “right”, “front”, “back”, “top”, “bottom”, “inside”, “outside”, “middle”, “vertical”, “horizontal”, “lateral”, “longitudinal” and the like indicating directions or positional relationships are based on orientations or positional relationships shown in the accompanying drawings. These terms are mainly used to better describe the present disclosure and the implementation manners thereof, and are not used to limit that the device, element or component must have such a specific orientation, or be constructed and operated in such a specific orientation.

[0060] Further, in addition to indicating an orientation or a location relationship, some of the terms may also indicate other meanings, for example, the term “upper” may also indicate some attachment relationship or connection relationship in some cases. For those skilled in the art, the specific meanings of these terms in the present disclosure may be understood according to specific situations.

[0061] In addition, the terms “install”, “arrange”, “provided with”, “form”, “connect” and “join” and the like shall be understood in a broad sense. For example, “connection” may refer to “fixed connection”, “detachable connection”, or “integral connection as a whole”; may refer to “mechanical connection”, or “electrical connection”; and may refer to “direct connection”, or “indirect connection through an intermediate medium”, or “internal communication between two devices, elements, or components. For those skilled in the art, the specific meaning of these terms in the present disclosure may be understood according to specific conditions.

[0062] In addition, the terms “first”, “second” and the like are mainly used to distinguish different devices, elements or components (specific types and configurations may be the same or different), and are not used to indicate or imply relative importance and quantity of the indicated devices, elements or components. Unless otherwise indicated, “a plurality of” means two or more.

[0063] The implementation details of the linear vibration motor and the electronic device of the present disclosure are described in details below, and the following contents are only implementation details provided for ease of understanding, and are not necessary for implementing the present solution.

[0064] Referring to FIG. 1 to FIG. 9, Embodiment I of the present disclosure provides a linear vibration motor **100**, including: a case **10**, a vibrator **20**, a driver **30**, and a plurality of support members **40**. The case **10** has a receiving space **11**, the vibrator **20** includes a support frame **21** and a follower **22** fixed to the support frame **21**, and the support frame **21** is located in the receiving space **11**. The driver **30** is arranged opposite to the follower **22**. The driver **30** is also located in the receiving space **11**, and the driver **30** is configured to drive the follower **22** to drive the support frame to move along a straight line L. The support frame **21** is provided with a plurality of

receiving grooves **211**, each of which is provided with at least one support member **40**, and a part of the support member **40** is located outside the receiving groove **211**. The support member **40** is pressed on the case **10** by the support frame **21**. When the driver **30** drives the support frame **21** to move, the support member **40** rolls relative to the support frame **21**.

[0065] In this way, when the driver **30** drives the follower **22** to make the support frame **21**

[0066] move along with the follower **22**, the support member **40** not only can support the support frame **21**, but also can allow the support frame **21** to move back and forth relative to itself along the straight line L, so that the linear vibration motor **100** can vibrate back and forth. Moreover, since the receiving groove **211** is located inside the receiving space **11** of the case **10**, it will not deform under an action of falling impact and the like, and normal movement of the support member **40** will not be affected. Thus, the linear vibration motor **100** has good impact resistance and vibration resistance.

[0067] In this embodiment, the case **10** includes a front shell **12** and a base **13**, the front shell **12** is of a groove structure, and the base **13** covers the groove structure to define a receiving space **11** together with the groove structure. For example, the front shell **12** is of a square groove structure, the base **13** is of a square plate structure, and the base **13** covers an opening of the square groove structure, to form the receiving space **11**. The front shell **12** and the base **13** may be fixed to each other by welding or bonding, so as to form an integrated structure, which may have excellent waterproof performance and dustproof performance. Further, the front shell **12** and the base **13** may be made of a material with high strength, so as to prevent deformation of the case **10** due to falling impact and other reasons.

[0068] Referring to FIG. 4, FIG. 5 and FIG. 7 again, in the embodiment, the linear vibration motor **100** further includes a second magnetic member **50** arranged at the front shell **12** or the base **13**. The driver **30** is a coil, and the coil is arranged at the front shell **12** or the base **13**. The follower **22** is a first magnetic member. The second magnetic member **50** is arranged opposite to the first magnetic member, and the second magnetic member **50** is configured to cooperate with the first magnetic member to make the support frame **21** maintain at an initial position when the coil is powered off, and drive the first magnetic member to drive the support frame **21** to move along the straight line L when the coil is powered on.

[0069] In addition, the second magnetic member **50** is further configured to make the support frame **21** have a tendency to move towards the case **10** (which may be the front shell **12** or the base **13**), so that the support frame **21** can press the support member **40** onto the case **10**. In some examples, the second magnetic member **50** arranged at the case **10** is attracted to the first magnetic member arranged at the support frame **21**, so that the support frame **21** has a tendency to move towards the front shell **12**, thus the front shell **12** and the support frame **21** jointly clamp and fix the ball.

[0070] For example, the second magnetic member **50** is arranged at the front shell **12** and the coil is arranged at the base **13**. The magnetic field of the second magnetic member **50** itself interacts with the magnetic field of the first magnetic member (such as magnetic steel) itself, and since the second magnetic member **50** is fixed at the front shell **12** and cannot move, the first magnetic member will move under an action of a force. In this embodiment, the second magnetic member **50** is configured to attract the first magnetic member, and the second magnetic member is arranged at a central position of the front shell **12**. In this way, in an initial state (when the coil is powered off), the second magnetic member **50** can attract the first magnetic member to maintain it in an initial position; and when the coil is powered on, the coil generates a magnetic field, and the magnetic field generated by the coil interacts with the magnetic field of the first magnetic member. Since the coil is fixed at the base **13** and does not move, when an acting force of the magnetic field generated by the coil on the first magnetic member is greater than an acting force of the magnetic field of the second magnetic member on the first magnetic member, the first magnetic member moves under the action of a force to deviate from the initial position, thereby driving the support frame **21** to move. By controlling the magnitude and the direction of the current passing through the coil, a

direction of the force applied to the first magnetic member can be controlled, so that the support frame **21** can move along an expected direction. When the support frame **21** moves back and forth, vibration of the linear vibration motor **100** is achieved. It should be noted that, when the coil drives the support frame **21** to move, the force of the second magnetic member **50** on the first magnetic member always exists, and the force makes the support frame **21** have a tendency to return to the initial position when the support frame **21** is going away from the initial position. That is, the second magnetic member **50** may serve as a magnetic spring. When the coil is powered off, the magnetic field generated by the coil disappears, so that the first magnetic member is no longer affected by the magnetic field generated by the coil. At this time, the first magnetic member moves to the initial position by the force of the magnetic field of the second magnetic member **50**.

[0071] In some examples, the second magnetic member **50** may be a magnetic yoke, or may be magnetic steel, or may be magnetic fluid encapsulated and fixed to the front shell **12** or the base **13** by using a seal member, as long as the second magnetic member **50** may serve as a magnetic spring when the coil is powered on and the support frame **21** is driven to move, or can make the first magnetic member maintain at an initial position when the coil is powered off.

[0072] Referring to FIG. **10**, it should be noted that, when the first magnetic member and the second magnetic member **50** are both made of magnetic steel, different magnetic poles of the first magnetic member and the second magnetic member **50** need to be oppositely arranged, for example, the N pole of the first magnetic member is opposite to the S pole of the second magnetic member **50**, and the S pole of the first magnetic member is opposite to the N pole of the second magnetic member **50**. In this way, opposite surfaces of the first magnetic member and the second magnetic member **50** can be attracted to each other.

[0073] It can be understood that by adjusting a specification size (such as length, width, and height) of the second magnetic member **50**, different acting forces of the second magnetic member **50** on the first magnetic member can be obtained, so that a spring coefficient having a small negative impact on the linear vibration motor **100** can be obtained in the receiving space **11** only by adjusting the second magnetic member **50**. Or, different forces of the second magnetic member **50** on the first magnetic member can be obtained through a combination of a plurality of second magnetic members **50**, and the spring coefficient can be changed only by changing a combination of the second magnetic members **50** compared with the solutions in the related art in which a leaf spring is used.

[0074] It can be understood that positions of the coil and the second magnetic member **50**

[0075] may be interchanged, that is, the coil is arranged at the front shell **12** and the second magnetic member **50** is arranged at the base **13**. Or, as shown in FIG. **11**, the coil and the second magnetic member **50** may be arranged at a same side, for example, the coil and the second magnetic member **50** are both arranged at the front shell **12**, or the coil and the second magnetic member **50** are both arranged at the base **13**. In this case, either of the coil and the second magnetic member **50** may be directly arranged at the front shell **12** or the base **13**, and the other is arranged at a side of the former away from the front shell **12** or the base **13**.

[0076] Referring to FIG. **9** to FIG. **11**, in an embodiment of the present disclosure, the receiving groove **211** is provided at a side of the support frame **21** facing the front shell **12**, that is, the receiving groove **211** opens towards or substantially opens towards the front shell **12**. The receiving groove **211** extends along a straight line L. An inner wall of the receiving groove **211** and an inner wall of the front shell **12** clamp and fix the support member **40**, and the support member **40** is a ball. In this way, in a working process of the linear vibration motor **100**, when the support frame **21** moves back and forth, the ball rolls inside the receiving groove **211**, so that the support frame **21** can move stably. In other words, as long as the ball rolls normally, a normal working of the linear vibration motor **100** is not affected relative to a specific position of the receiving groove **211** or the front shell **12**. Therefore, the linear vibration motor **100** provided by this embodiment of the present disclosure is not easily affected by the assembly precision.

[0077] It can be understood that, when the second magnetic member **50** is arranged at the front shell **12**, in a drop test or an unexpected drop impact, the attraction between the first magnetic member and the second magnetic member **50** may also reduce possible collision between the support frame **21** and the base **13**, and prevent dust generated due to abrasion of the support frame **21** and/or the base **13** caused by friction when the support frame **21** and the base **13** collide.

[0078] In an example, the support member **40** is a roller, and the support member **40** is arranged at a side of the support frame **21** facing the front shell **12**. The wheel axle of the roller is fixed to the support frame **21**, the axial direction of the wheel axle is parallel to the base **13**, and the roller may rotate by taking the axis of the wheel axle as a rotation axis. When the support frame **21** moves, the roller rolls at the front shell **12**.

[0079] In this embodiment, at least three receiving grooves **211** are provided, and at least one receiving groove **211** is arranged at a position that is not in the same straight line as the positions of the other receiving grooves **211**. For example, exactly three receiving grooves **211** are provided, and the positions of the three receiving grooves **211** are in a same plane and form a triangle, so that the support frame **21** does not flip over relative to the front shell **12**.

[0080] In this embodiment, the support frame **21** is of a square ring shape, four receiving grooves **211** are provided. The four receiving grooves **211** are located at four corners of the support frame **21**, and a respective one ball is provided at each of at least three receiving grooves **211**. In other embodiments, more than four receiving grooves **211** may be provided.

[0081] In an example, the inner wall of the front shell **12** includes a front wall **121** opposite to the support frame **21**, and a side wall **122** surrounding the front wall **121** and connected to the front wall **121**. The inner wall of the receiving groove **211** includes an inclined wall **2111** which forms an included angle with the front wall **121**. The balls are located between the front wall **121**, the side wall **122**, and the inclined wall **2111**. Referring to FIG. 8, taking the ball located at an upper position shown in FIG. 7 as an example, under an action of the attraction of the first magnetic member and the second magnetic member **50**, the inclined wall **2111** applies an inclined upper left force **F1** to the ball, the front wall **121** applies a horizontal right force **F2** to the ball, and the side wall **122** applies a vertical downward force **F3** to the ball. When the support frame **21** is in a static state, the effects of these forces reach balance, and the position of the ball is maintained. Under the action of these forces, when the support frame **21** is driven to move, the balls may only roll along the straight line **L**. Since the support frame **21** is supported by the ball, the support frame **21** may only move along the direction of the straight line **L**. For the ball located at a lower position, the inclined wall **2111** of the corresponding receiving groove **211** applies an inclined downward left force, and the side wall **122** corresponding to this position applies a vertical upward force.

[0082] In some examples, the included angle between the inclined wall **2111** and the front wall **121** is within a range from 30° to 60°, for example, 45°.

[0083] Referring to FIG. 6, in this embodiment, the receiving groove **211** further includes a bottom wall **2112** and an upper wall **2113** both connected to the inclined wall **2111**. The bottom wall **2112** is perpendicular to the front wall **121**, and the upper wall **2113** is parallel to the front wall **121**. In some cases, both the bottom wall **2112** and the upper wall **2113** may limit the position of the ball, ensuring that the ball is at a correct position.

[0084] Referring to FIG. 4 to FIG. 6, in an example, the support frame **21** is provided with an anti-collision portion **212** at a side away from the receiving groove **211**, and in a direction perpendicular to the base **13**, a width of a gap between the anti-collision portion **212** and the base **13** is less than a height of the support member **40**. The anti-collision portion **212** is configured to prevent the support frame **21** from colliding with the base **13**. In a drop test or an unexpected drop impact, the anti-collision portion **212** may abut against the base **13** before the support frame **21** contacts the coil or the base **13**, thereby avoiding damage to the support frame **21** or preventing the first magnetic member from disengaging from the support frame **21** due to impact.

[0085] The width of the gap between the anti-collision portion **212** and the base **13** is smaller than a

diameter of the ball. Even if the anti-collision portion **212** is in full contact with the base **13**, the width of the gap between the support frame **21** and the front shell **12** is smaller than the diameter of the ball, which may prevent the ball from separating from the receiving groove **211**. In this way, the action of the falling impact is slowed down, or after the falling impact is finished, under an action of the first magnetic member and the second magnetic member **50**, the support frame **21** is back to an initial position and the ball moves back to the inclined wall **2111**.

[0086] It should be noted that when the first magnetic member is arranged at the support frame **21**, and the coil and the anti-collision portion **212** are both located at a same side of the support frame **21**, in a direction perpendicular to the base **13**, the height of the anti-collision portion **212** should be greater than the height of the coil. In this case, when the anti-collision portion **212** contacts the base **13** due to the falling impact, the support frame **21** or the first magnetic member does not contact the coil. If the coil and the anti-collision portion **212** are arranged at two opposite sides (that is, not at a same side) of the support frame **21**, the support frame **21** may be provided with an avoidance groove **213** for avoiding the coil. When the support frame **21** firmly presses the ball, a distance between a bottom of the avoidance groove **213** and the case **10** should be greater than the height of the coil.

[0087] In order to improve the impact resistance of the anti-collision portion **212**, the anti-collision portion **212** may be formed by a material having certain elasticity, such as rubber or other material, and the anti-collision portion **212** may also be integrally formed with the support frame **21**.

[0088] During the working process of the linear vibration motor **100**, the attraction between the first magnetic member and the second magnetic member **50** may also cause a certain gap between the anti-collision portion **212** and the base **13**, so that the linear vibration motor **100** may continuously and stably output.

[0089] Referring to FIG. **12** and FIG. **13**, in other examples, the receiving groove **211** is arranged at a side of the support frame **21** facing the base **13** and extends along the straight line L, and the inner wall of the receiving groove **211** and the base **13** clamp and fix the ball. In this case, since the support frame **21** and the base **13** need to jointly clamp and fix the ball, the second magnetic member **50** may be arranged at the base **13**, for example, at a center position of the base **13**. In this case, the ball is located between the base **13**, the side wall **122** and the inclined wall **2111**. In this case, the base **13** applies a horizontal leftward force to the ball, the side wall **122** applies a vertical downward force to the ball, and the inclined wall **2111** applies an inclined rightward force to the ball. When these forces are balanced, the position of the ball is maintained.

[0090] It can be understood that when the receiving groove **211** opens towards the base **13**, the anti-collision portion **212** is arranged at a side of the support frame **21** facing the front shell **12**.

[0091] Referring to FIG. **5**, in an example, the linear vibration motor **100** further includes a circuit board **60**. The circuit board **60** is electrically connected to the coil to supply power to the coil. When the coil is fixed to the front shell **12**, the circuit board **60** may also be fixed to the front shell **12**; and when the coil is fixed to the base **13**, the circuit board **60** may also be fixed to the base **13**.

[0092] Further, in order to connect the circuit board **60** from the outside of the case **10**, a connection end of the circuit board **60** may be exposed to the outside of the case **10**, so that a probe with a spring may be used for contact connection, or a solder connection may be used, or an ACF (Anisotropic Conductive Film) may be used for electrical connection to the controller of the electronic device.

[0093] Referring to FIG. **13** and FIG. **14**, Embodiment II of the present disclosure provides a linear vibration motor **100**, which is substantially the same as the linear vibration motor **100** of Embodiment I, and its main difference is as follows: the second magnetic member **50** is arranged at the support frame **21**, the driver **30** is the first magnetic member, and the follower **22** is a coil. When the coil is powered off, the second magnetic member **50** is attracted by the first magnetic member, so that the support frame **21** is maintained at the initial position; and when the coil is powered on, the magnetic field generated by the coil interacts with the first magnetic member, and

because the position of the first magnetic member is fixed, the coil drives the support frame **21** to move along the straight line L to deviate from the initial position.

[0094] In this case, the second magnetic member **50** may be located between the follower **22** and the driver **30**, or may be located at a side of the follower **22** away from the driver **30**. In some examples, the second magnetic member is arranged at a side of the follower **22** away from the driver **30**, so that a certain gap may be maintained between the vibrator **20** and the driver **30**, thereby avoiding contact between the vibrator **20** and the driver **30** during impact.

[0095] Since the first magnetic member is fixed to the case **10**, the height of the anti-collision portion **212** should be greater than the height of the first magnetic member; or when the support frame **21** firmly presses the ball, a distance between the bottom of the avoidance groove **213** at the support frame **21** and the case **10** should be greater than the height of the first magnetic member.

[0096] In this case, since the coil will move continuously when the linear vibration motor **100** works, the circuit board **60** is at least partially configured to be flexible to deform in cooperation with the movement of the coil.

[0097] According to the above-described embodiments, it can be found that either of the first magnetic member and the coil can be arranged at the support frame **21**, and the other one is arranged at the case **10**, and the specific arrangement manner is not specifically limited herein. The same is true for the first magnetic member and the second magnetic member **50**, either of which is arranged at the support frame **21** and the other one is arranged at the case **10**. The magnetic member arranged at the case **10** needs to be disposed opposite to the opening of the receiving groove **211**, to ensure that the case **10** and the support frame **21** can clamp the ball.

[0098] For a traditional leaf spring structure, in a drop test or an unexpected drop impact, the shock that cannot be absorbed causes the vibrator **20** to vibrate vigorously relative to the case **10**, and this vibration has a large damage to the components of the vibrator **20** and other components fixed to the case **10**, also, the leaf spring structure is prone to damage due to stress concentration and the like. In the embodiments of the present disclosure, a traditional leaf spring structure is omitted, and the design of the magnetic spring is achieved by the attraction effect between the first magnetic member and the second magnetic member **50**. In this way, the reliability reduction and spring breakage caused by metal fatigue can be avoided, and the spring coefficient of the magnetic spring can be changed by adjusting the length, width and height of the second magnetic member **50** to achieve various changes, thereby reducing the design difficulty and reducing the limiting factors for designing the linear vibration motor **100**. Moreover, the attraction between the first magnetic member and the second magnetic member **50** makes the vibrator **20** not always contact the case **10**, and even in a drop test and an unexpected drop impact, this attraction may reduce the vibration of the vibrator **20**, and the sliding support of the ball on the support frame **21** can also reduce the negative effect caused by the impact, thereby improving the impact resistance and vibration resistance of the linear vibration motor **100**.

[0099] In addition, for a conventional leaf spring structure, the assembly precision of the linear vibration motor, a posture or state of the handheld electronic device, and a state of falling impact may cause contact (or collision) between the vibrator and the case (or stator). In the embodiments of the present disclosure, the vibrator **20** is slidably supported by the ball, and is continuously fixed by attraction of the first magnetic member and the second magnetic member **50**, so that a certain gap can be maintained between the vibrator **20** and the case **10** (or the stator). Moreover, a side of the support frame **21** away from the ball is also provided with an anti-collision portion **212**, and even in a case of accidental falling, the anti-collision portion **212** will first contact the case **10** to prevent the vibrator **20** from contacting the case **10** (or the stator).

[0100] Referring to FIG. 15 to FIG. 17, Embodiment III of the present disclosure provides an electronic device **200**, including a device body **210** and the linear vibration motor **100** described in Embodiment I or Embodiment II, and the linear vibration motor **100** is arranged at the device body **210**. Moreover, the number and specific positions of the linear vibration motors **100** provided at the

device body **210** may be selected and configured according to actual requirements.

[0101] In this embodiment, the electronic device **200** may be a portable electronic device such as a mobile phone or a tablet computer, or may be an electronic device using a linear vibration motor as a feedback device such as an automobile navigation system or a vehicle-mounted instrument panel, or may be a plurality of electronic devices including tactile feedback in a wearable device.

[0102] The linear vibration motor and the electronic device provided by the embodiments of the present disclosure are described in details in the above description, the principles and the embodiments of the present disclosure are described herein by using specific examples, and the description of the above embodiments is merely used to help understand the idea of the present disclosure, and there will be changes in the specific embodiments and application ranges. In summary, the contents of the present disclosure should not be construed as limiting the present disclosure.

Claims

1. A linear vibration motor, comprising: a case having a receiving space; a vibrator comprising a support frame and a follower fixed to the support frame, wherein the support frame is arranged in the receiving space, and the support frame is provided with a plurality of receiving grooves; a driver provided opposite to the follower, wherein the driver is arranged in the receiving space, and the driver is configured to drive the follower to drive the support frame to move along a straight line; and a plurality of support members, wherein each of the plurality of receiving grooves is provided with at least one of the plurality of support members, a part of one of the plurality of support members is located outside the receiving groove, the plurality of support frames press the plurality of support members against the case; and when the driver drives the support frame to move relative to the case, the support member rolls relative to the case and the support frame.
2. The linear vibration motor as described in claim 1, wherein the linear vibration motor further comprises a second magnetic member arranged at the case, the driver is a coil, and the coil is arranged at the case; the follower is a first magnetic member, the second magnetic member is disposed opposite to the first magnetic member, and the second magnetic member is configured to cooperate with the first magnetic member to maintain the support frame at an initial position when the coil is powered off, and drive the first magnetic member to drive the support frame to move along the straight line when the coil is powered on; or wherein the linear vibration motor further comprises a second magnetic member arranged at the support frame, the driver is a first magnetic member, and the first magnetic member is arranged at the case; the follower is a coil, the second magnetic member is disposed opposite to the first magnetic member, and the second magnetic member is configured to cooperate with the first magnetic member to maintain the support frame at an initial position when the coil is powered off, and the second magnetic member is further configured to be driven to drive the support frame to move along the straight line when the coil is powered on.
3. The linear vibration motor as described in claim 2, wherein the case comprises a front shell and a base, the front shell is of a groove structure, and the base covers the groove structure to enclose the receiving space with the groove structure; and the receiving groove is arranged at a side of the support frame facing the front shell and extends along the straight line, an inner wall of the receiving groove and an inner wall of the front shell clamp and fix the support member, and the support member is a ball.
4. The linear vibration motor as described in claim 3, wherein the inner wall of the front shell comprises a front wall opposite to the support frame, and a side wall surrounding the front wall and connected to the front wall; the inner wall of the receiving groove comprises an inclined wall which forms an included angle with the front wall; and the ball is located between the front wall, the side wall and the inclined wall.

5. The linear vibration motor as described in claim 4, wherein the included angle between the inclined wall and the front wall is within a range from 30° to 60°.
6. The linear vibration motor as described in claim 3, wherein an anti-collision portion is provided at a side of the support frame away from the receiving groove, and in a direction perpendicular to the base, a width of a gap between the anti-collision portion and the base is less than a height of the support member; and the anti-collision portion is configured to prevent the support frame from colliding with the base.
7. The linear vibration motor as described in claim 2, wherein the case comprises a front shell and a base, the front shell is a groove structure, the base covers the groove structure to enclose the receiving space with the groove structure; and the receiving groove is arranged at a side of the support frame facing the base and extends along the straight line, an inner wall of the receiving groove and the base clamp and fix the support member, and the support member is a ball.
8. The linear vibration motor as described in claim 7, wherein the front shell comprises a side wall surrounding the base and connected to the base, an inner wall of the receiving groove comprises an inclined wall which forms an included angle with the base, and the ball is located between the base, the side wall and the inclined wall.
9. The linear vibration motor as described in claim 8, wherein the included angle between the inclined wall and the base is within a range from 30° to 60°.
10. The linear vibration motor as described in claim 7, wherein an anti-collision portion is provided at a side of the support frame away from the receiving groove, and in a direction perpendicular to the base, a width of a gap between the anti-collision portion and the base is less than a height of the support member; and the anti-collision portion is configured to prevent the support frame from colliding with the front shell.
11. The linear vibration motor as described in claim 2, wherein the second magnetic member is a magnetic yoke.
12. The linear vibration motor as described in claim 2, wherein the second magnetic member is magnetic steel, and a magnetic pole of the second magnetic member is arranged opposite to a magnetic pole of the first magnetic member.
13. An electronic device, comprising a device body and a linear vibration motor arranged at the device body, wherein the linear vibration motor comprises: a case having a receiving space; a vibrator comprising a support frame and a follower fixed to the support frame, wherein the support frame is arranged in the receiving space, and the support frame is provided with a plurality of receiving grooves; a driver provided opposite to the follower, wherein the driver is arranged in the receiving space, and the driver is configured to drive the follower to drive the support frame to move along a straight line; and a plurality of support members, wherein each of the plurality of receiving grooves is provided with at least one of the plurality of support members, a part of one of the plurality of support members is located outside the receiving groove, the plurality of support frames press the plurality of support members against the case; and when the driver drives the support frame to move relative to the case, the support member rolls relative to the case and the support frame.
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